



EK-RA8D2

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Rev	Comment	Date	Drawn By
1.0	MP release	10 Jun 2025	SDD/KSS
1.01	Updated U10	16 Sep 2025	SDD/KSS
1.02	Updated X2	18 Nov 2025	SDD/KSS
0.0	Release notes	dd m..m yyyy	
0.0	Release notes	dd m..m yyyy	
0.0	Release notes	dd m..m yyyy	
0.0	Release notes	dd m..m yyyy	
0.0	Release notes	dd m..m yyyy	
0.0	Release notes	dd m..m yyyy	

ERC Supression Markers

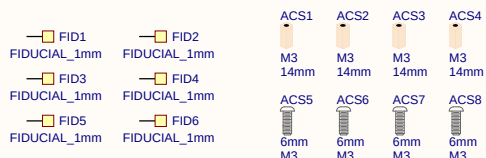
- ☒ General supress all ERC errors
- ☒ Unconnected Pins and Ports
- ☒ Floating Input Pin
- ☒ Power Nets with Power Pins and I/O objects
- ☒ Nets connected to multiple inputs
- ☒ Nets connected to Outputs and Bi-directional objects
- ☒ Nets connected to power nets and output objects
- ☒ Power pins connected to high impedance pins
- ☒ Nets connected to multiple outputs
- ☒ Net has no driving source

Note:

Labels shown inside parentheses:

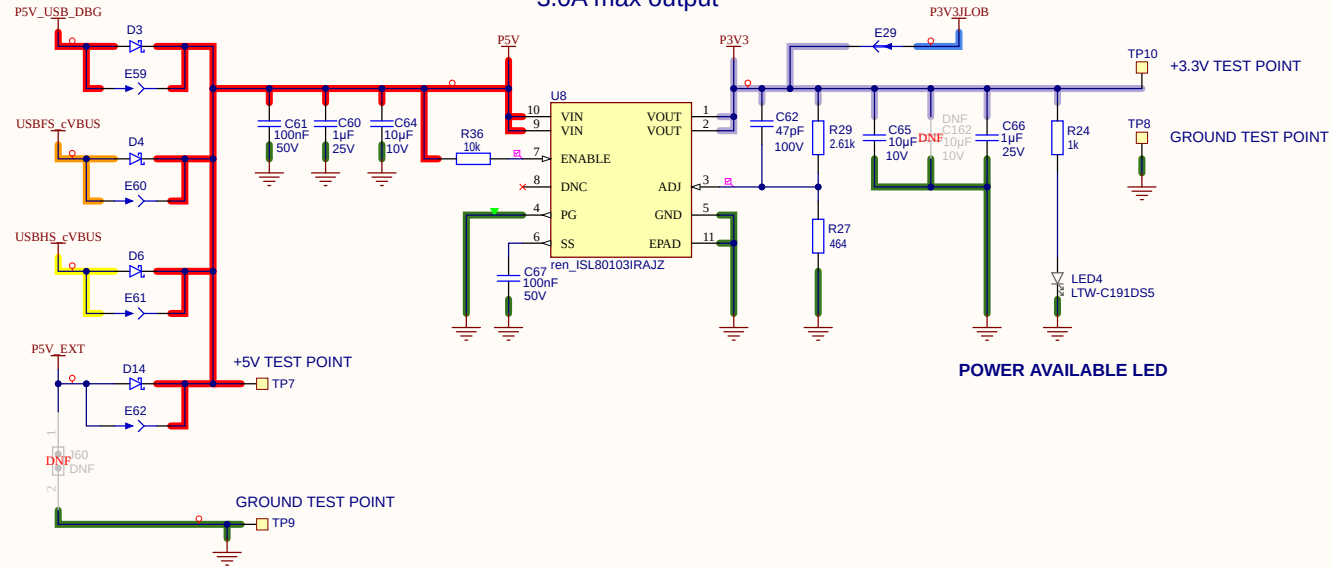
(note)

are NOT net names. They are text notes added to clarify signal usage.

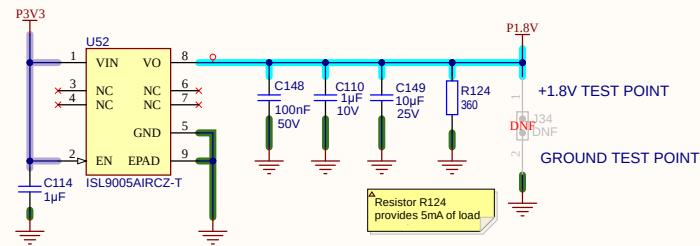


Power Supply

3.3V LINEAR REGULATOR 3.0A max output

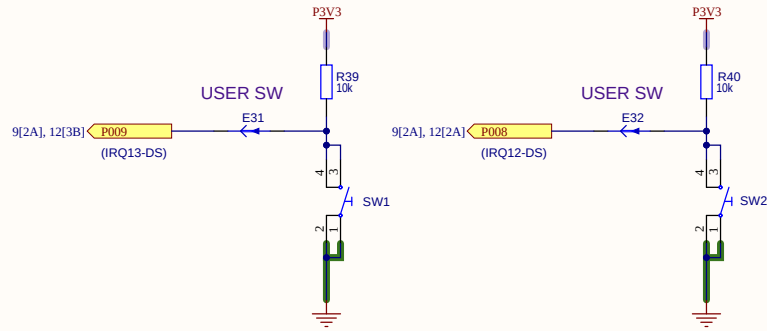


1.8V LINEAR REGULATOR 300mA max output

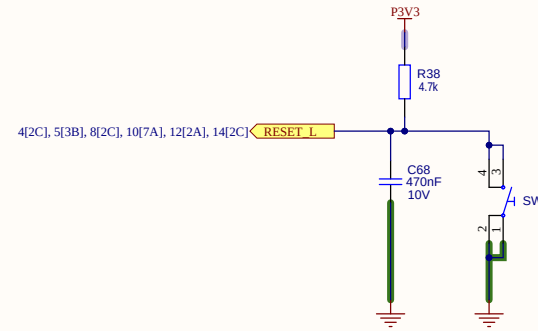


HMI Features

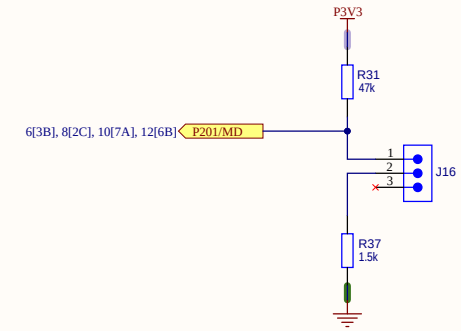
USER PUSH-BUTTONS



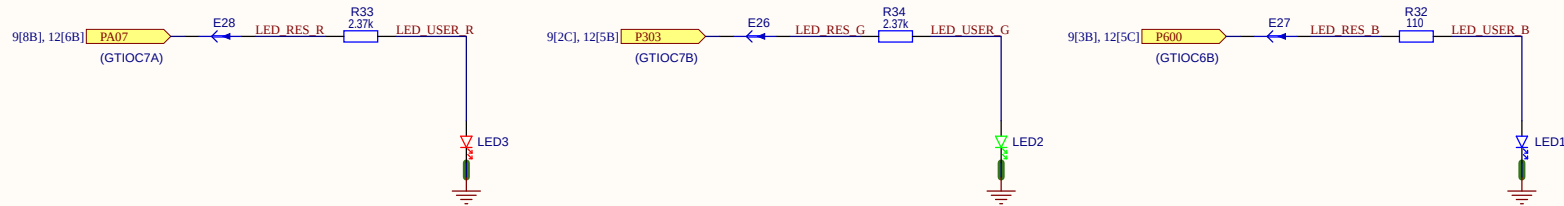
RESET PUSH-BUTTON



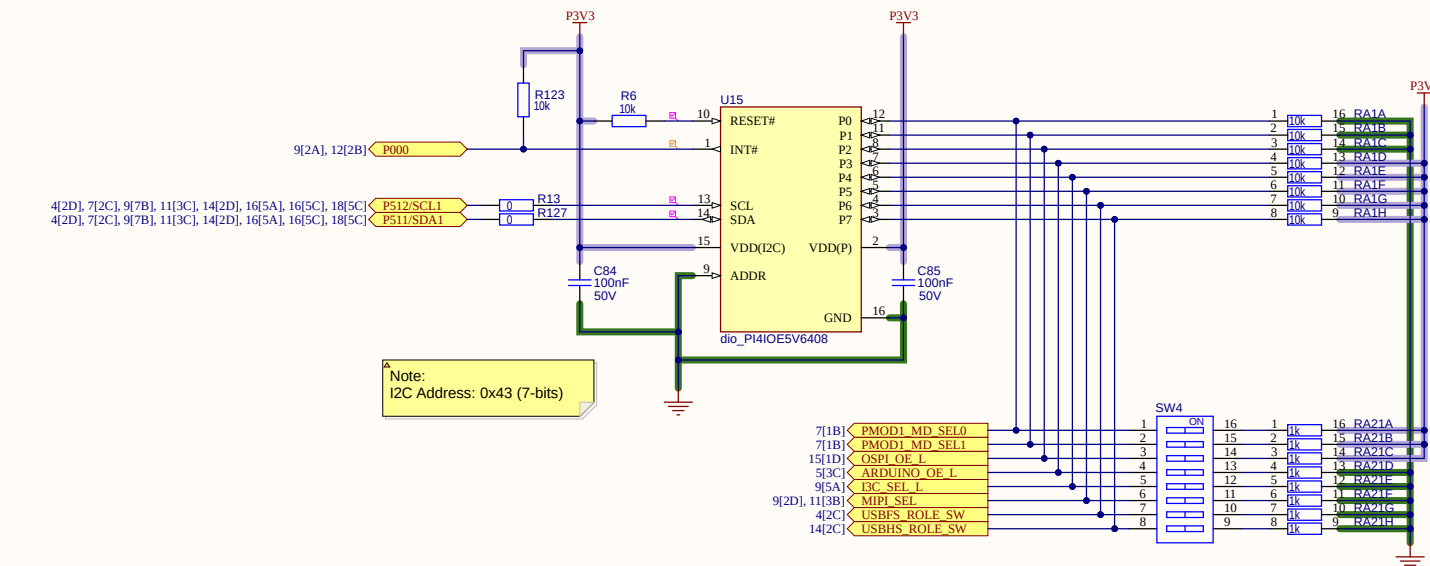
MCU BOOT MODE



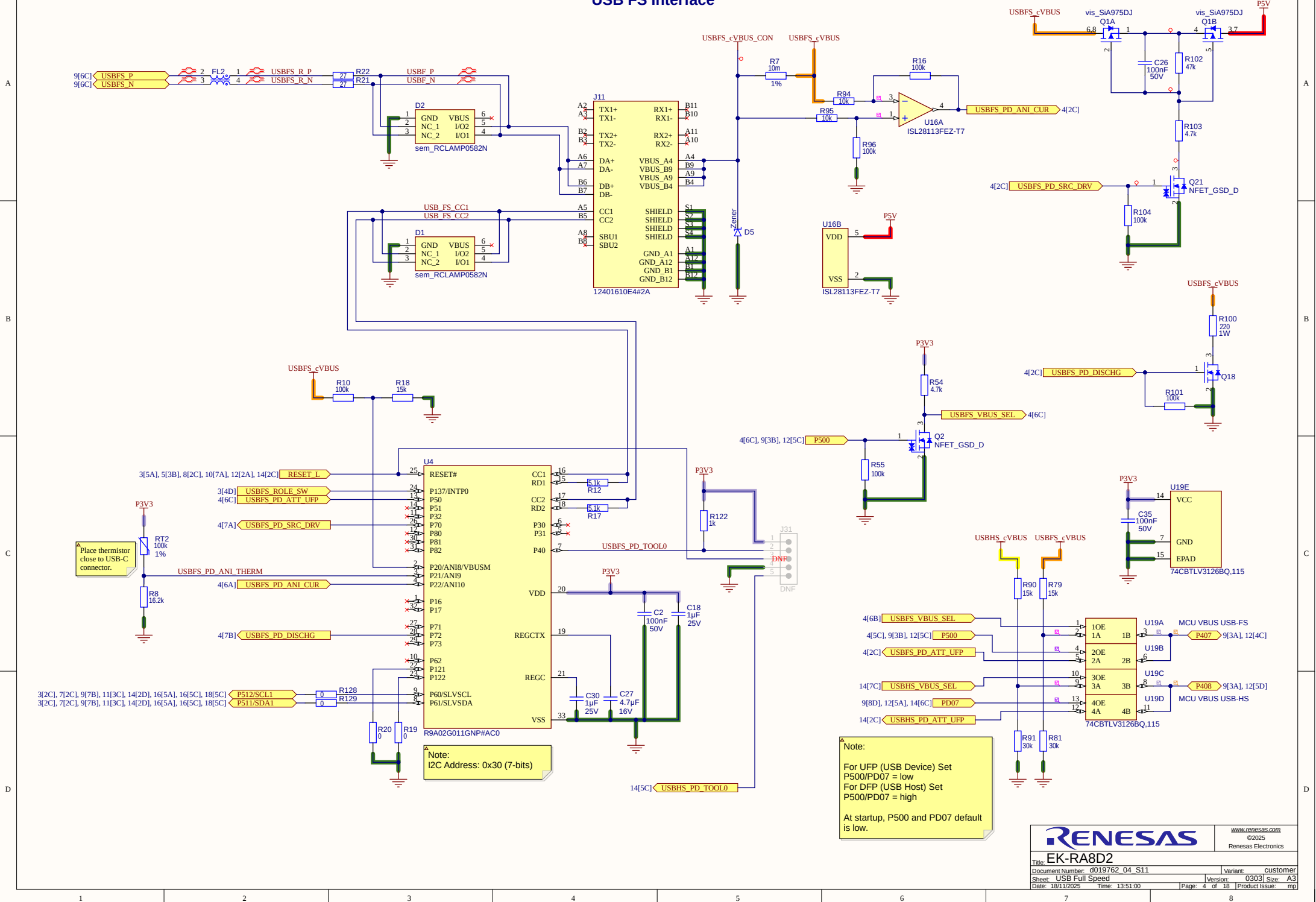
USER LEDs



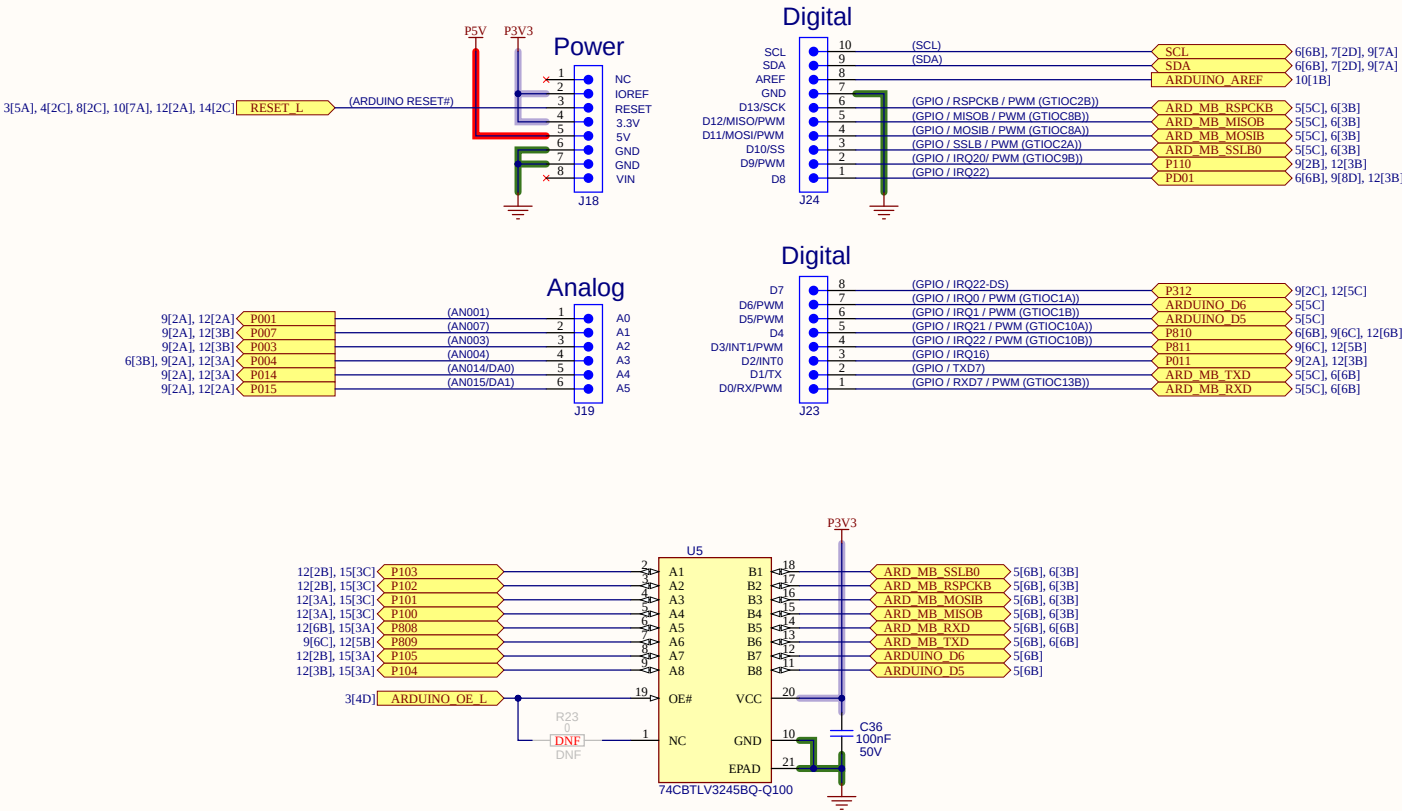
CONFIGURATION SWITCHES



USB FS Interface



Arduino Uno



A

B

C

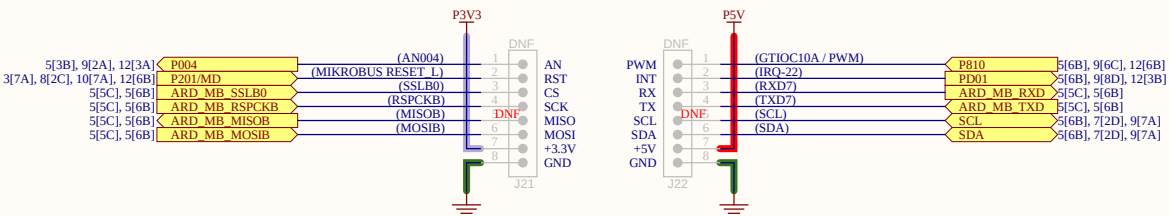
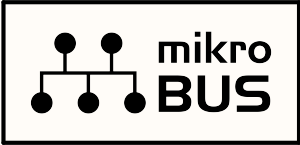
D

A

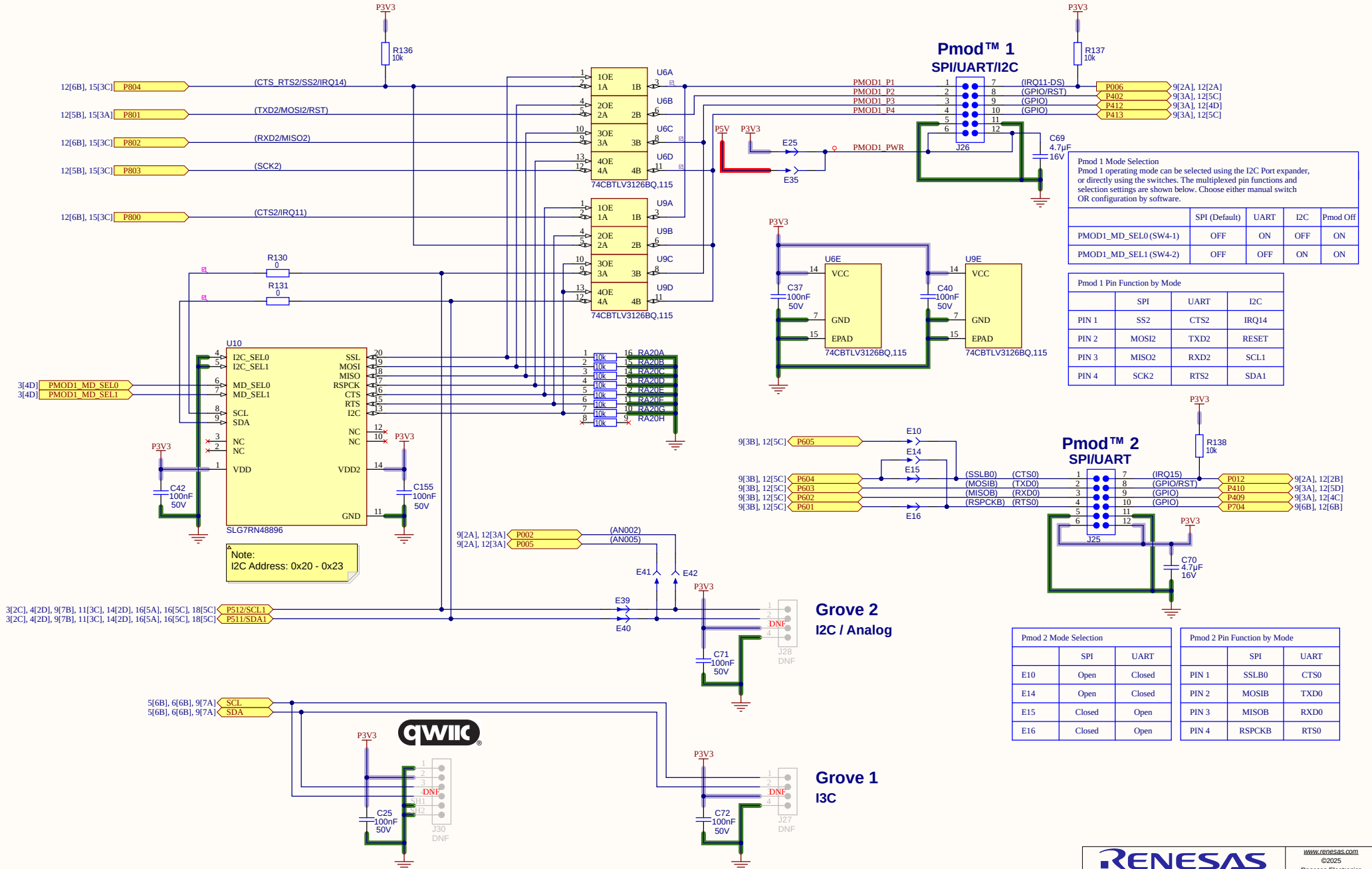
B

C

D

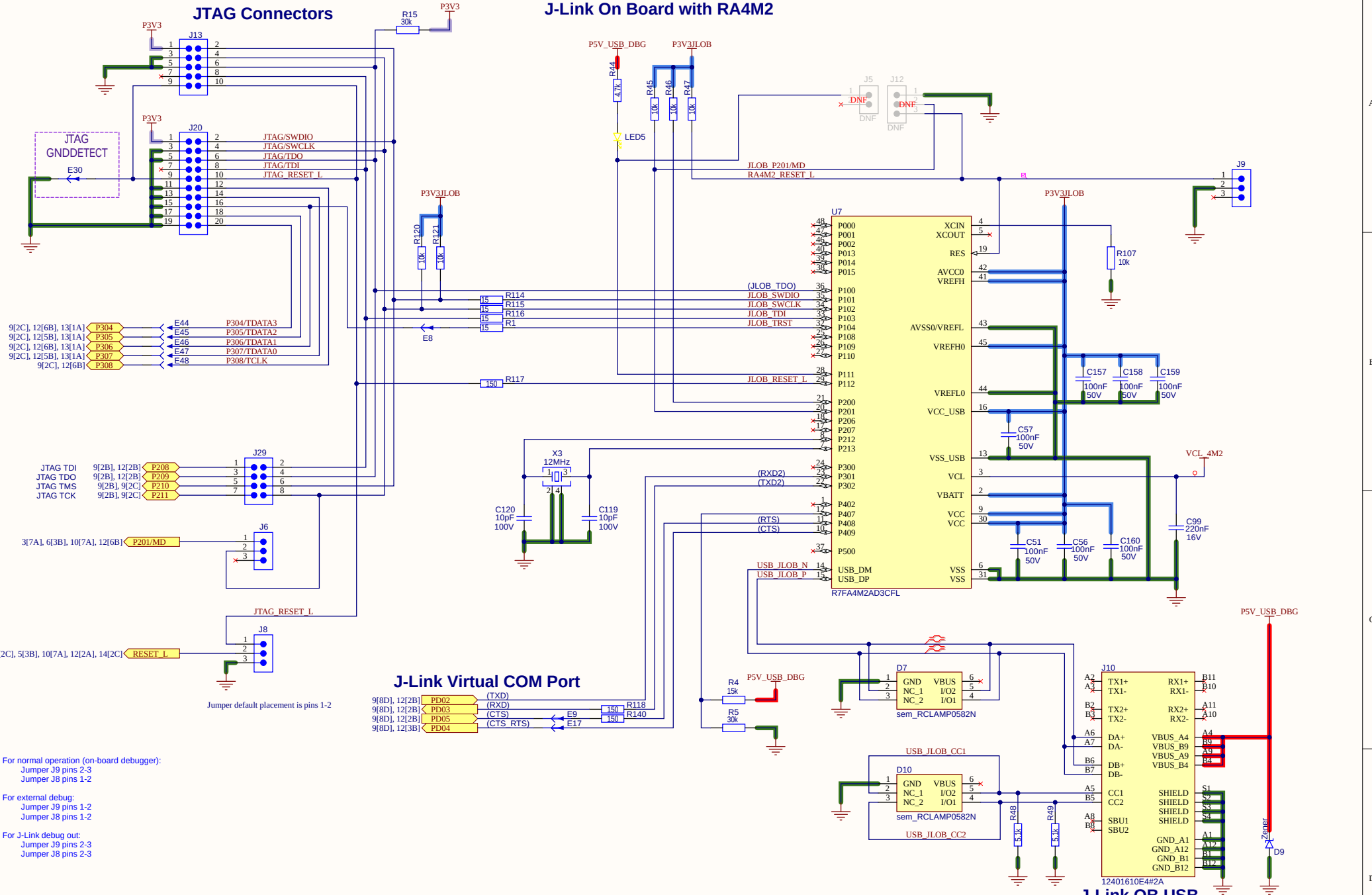


Pmod, Grove and Qwiic



JTAG Connectors

J-Link On Board with RA4M2

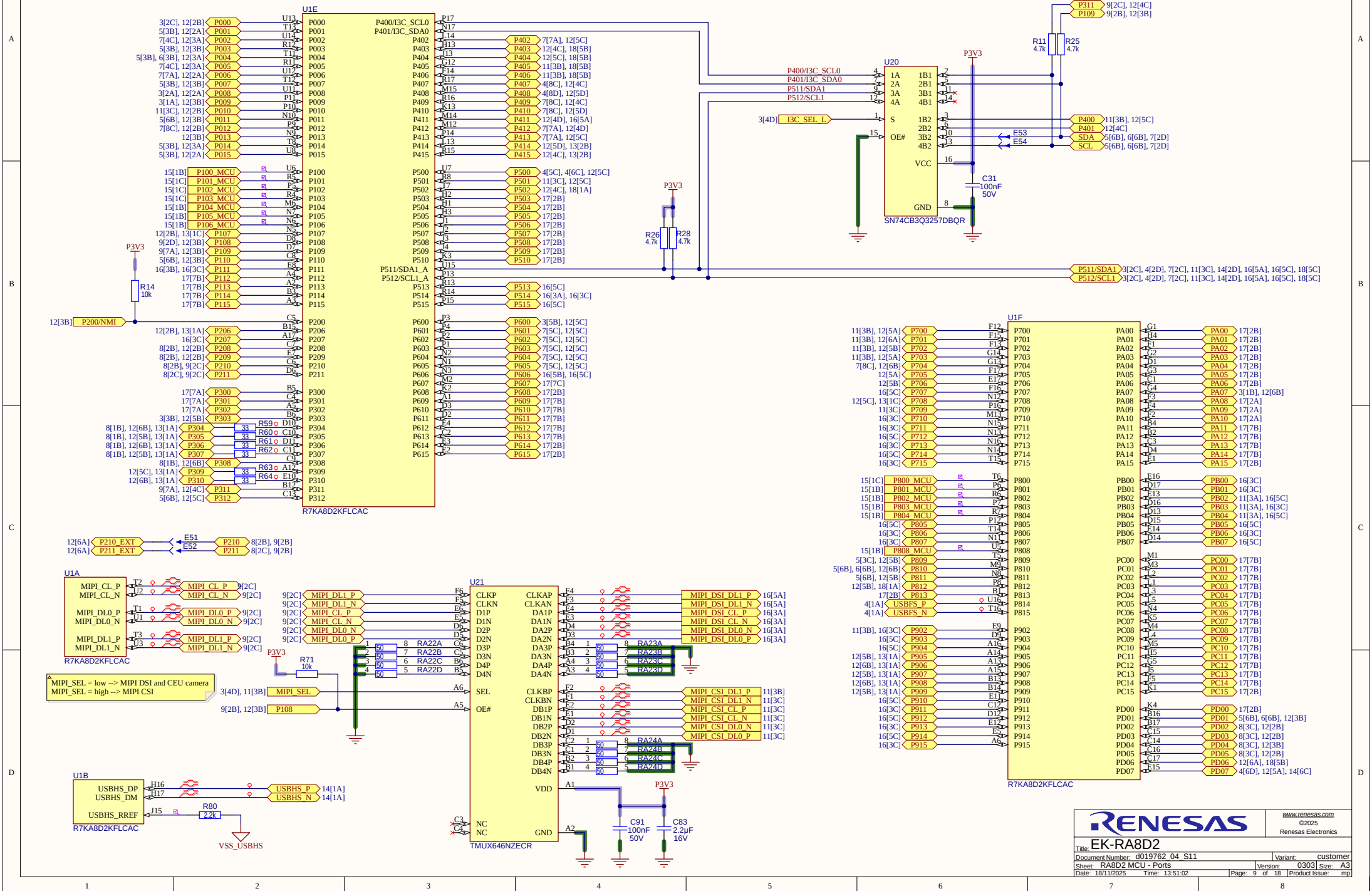


For normal operation (on-board debugger):
Jumper J9 pins 2-3
Jumper J8 pins 1-2

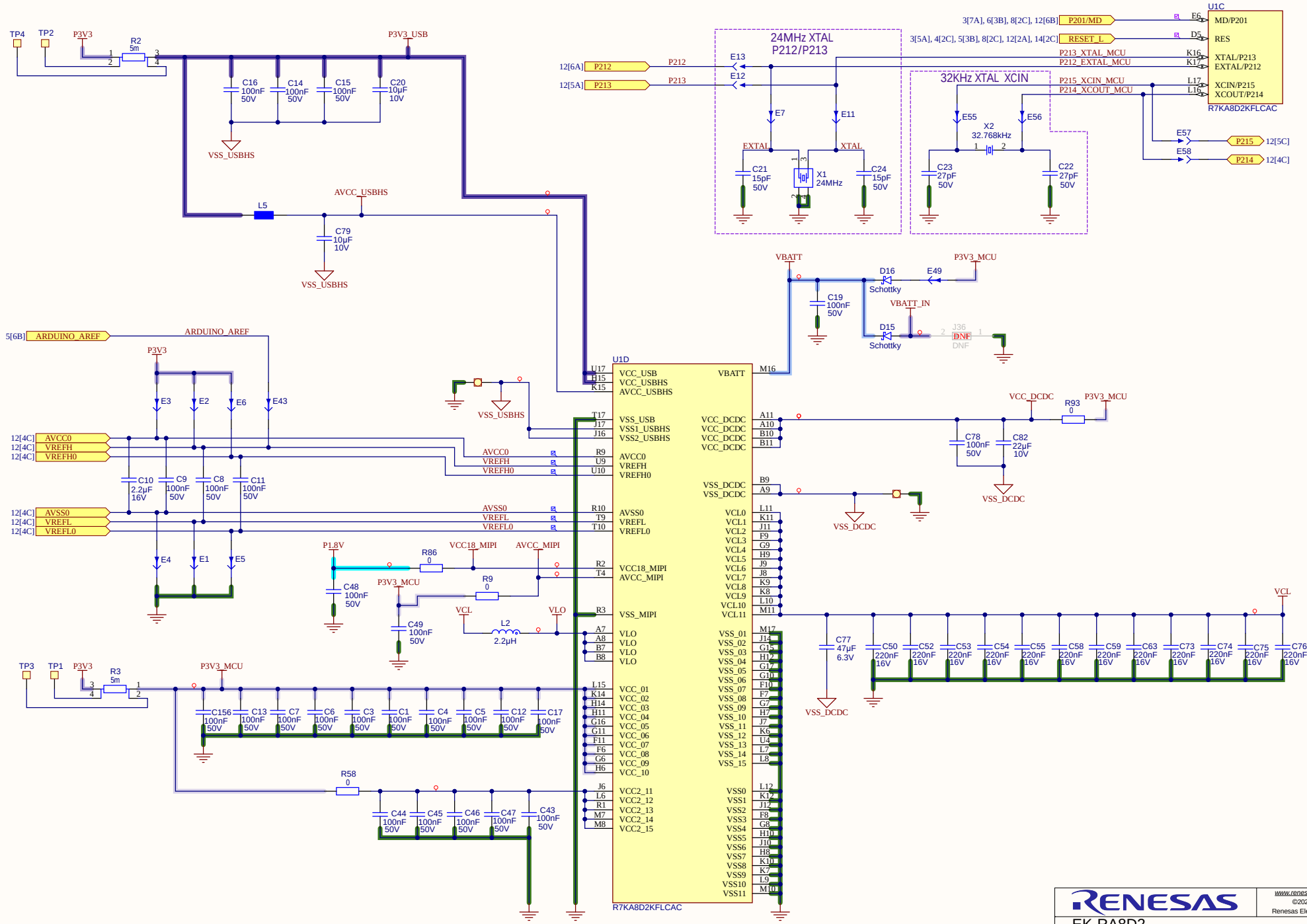
For external debug:
Jumper J9 pins 1-2
Jumper J8 pins 1-2

For J-Link debug out:
Jumper J9 pins 2-3
Jumper J8 pins 2-3

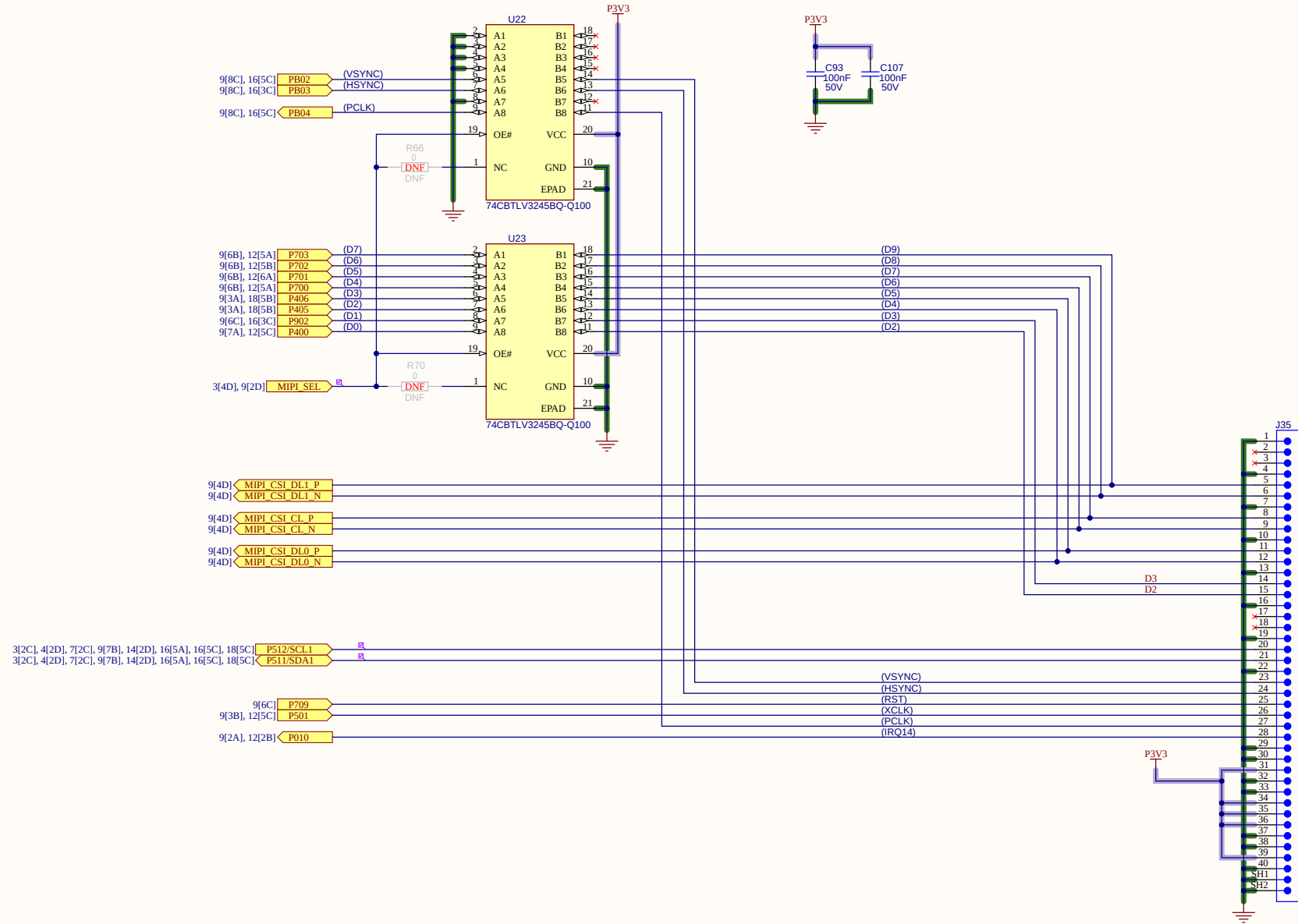
RA8D2 Microcontroller



RA8D2 Microcontroller



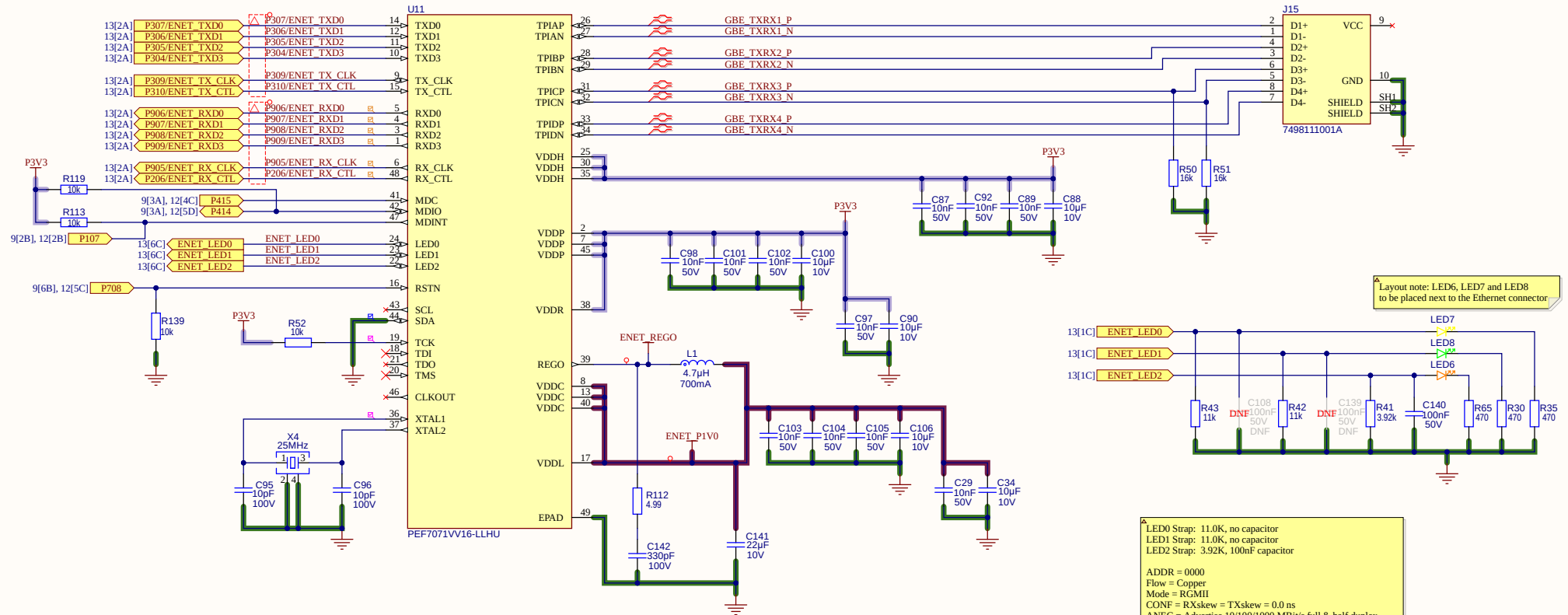
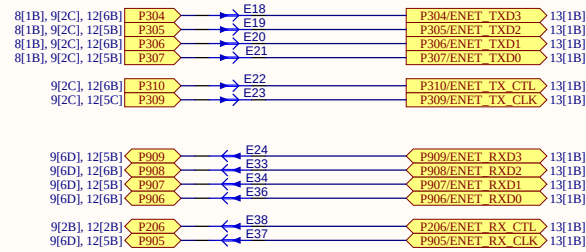
1	2	3	4
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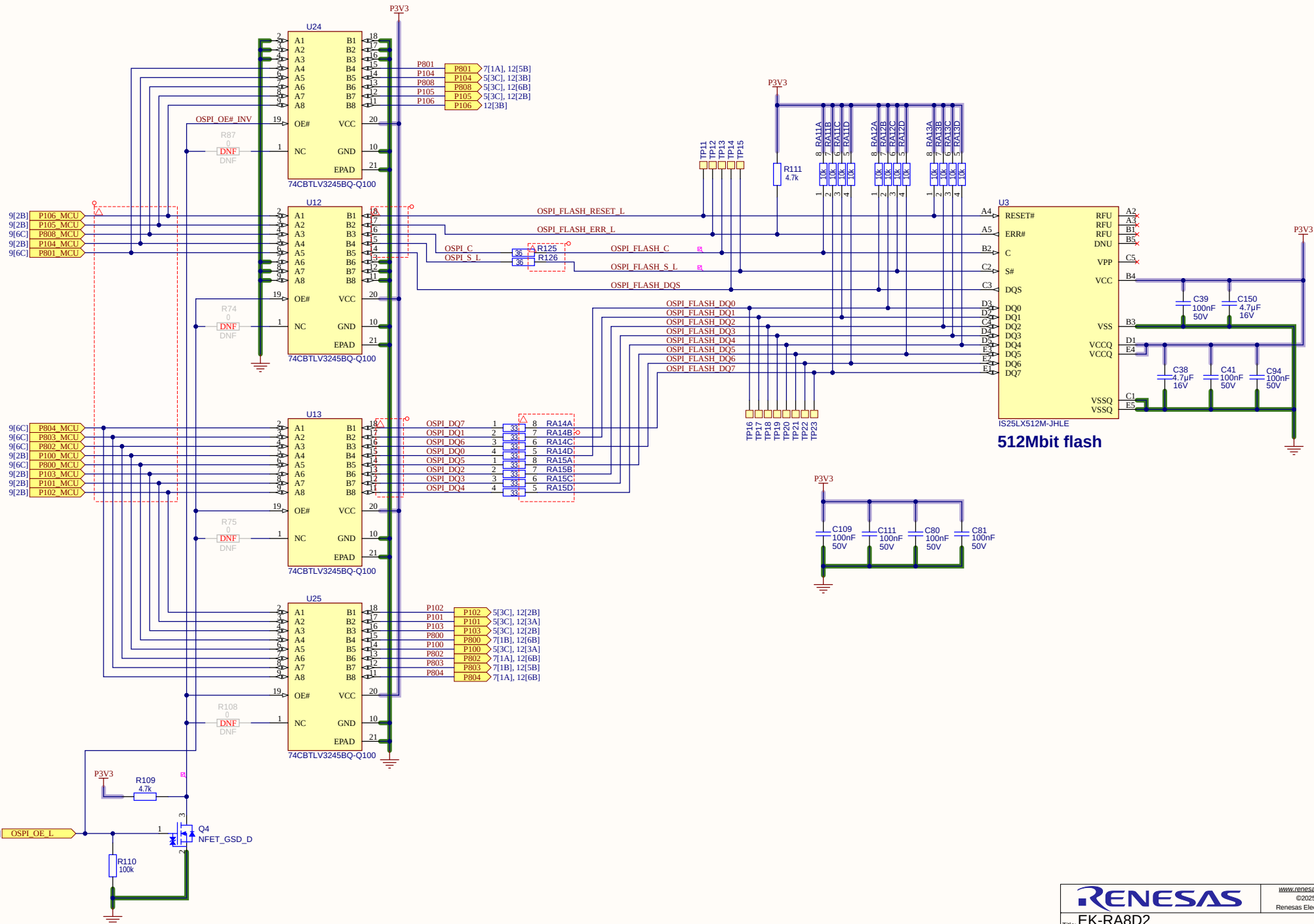
A



Gigabit Ethernet

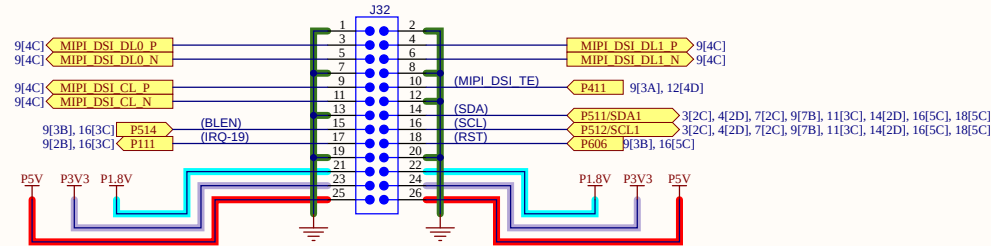


OCTAL SPI Flash

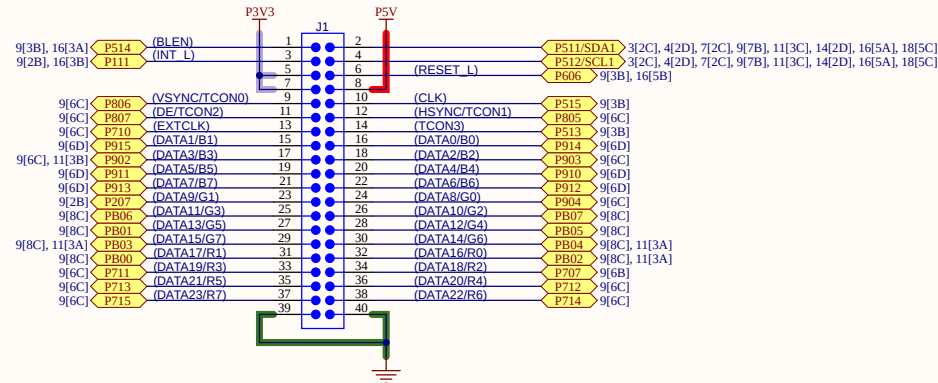


MIPI and Parallel Graphics Display Interface

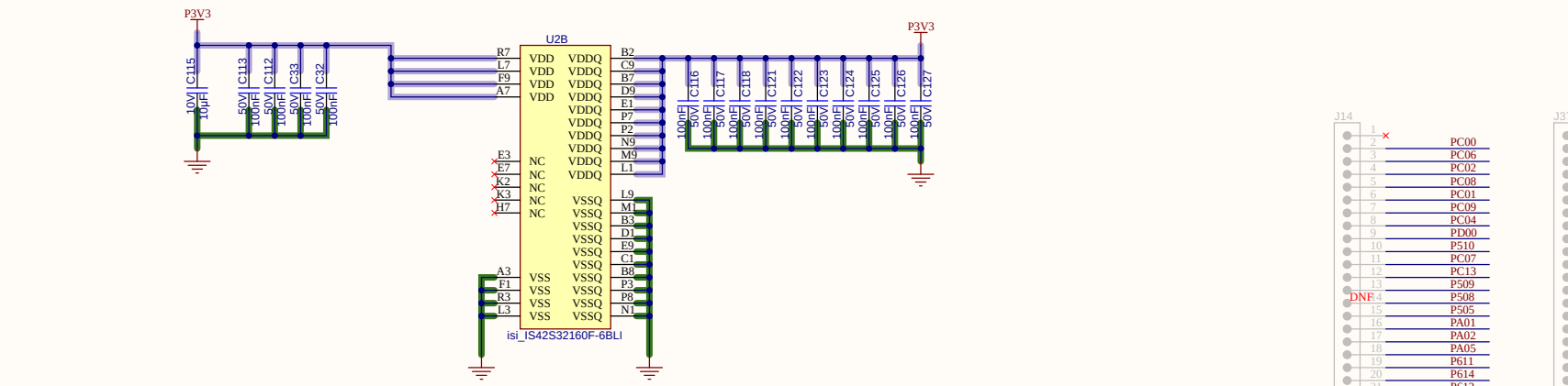
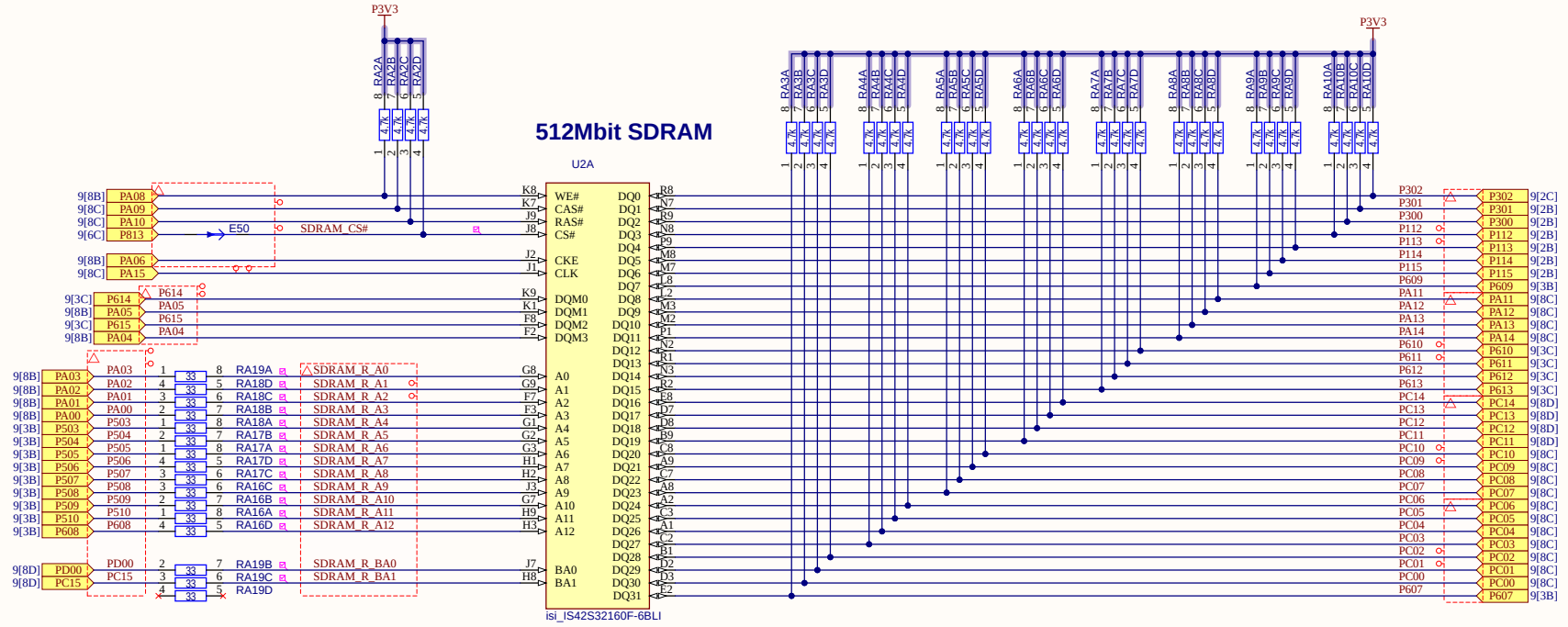
MIPI Connector



Parallel Graphics Display Connector



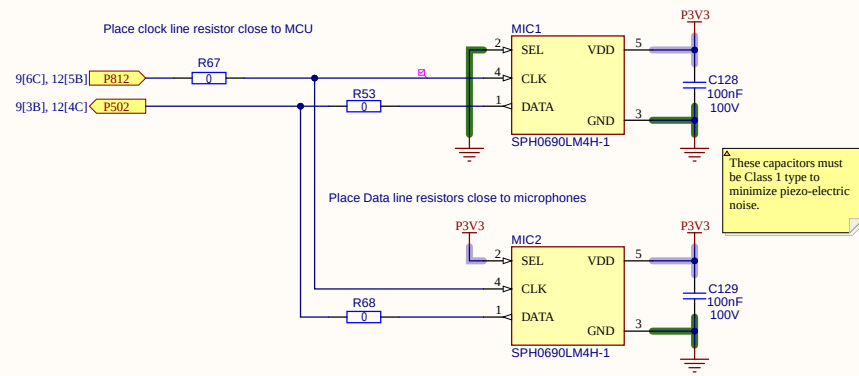
SDRAM



1	PC00
2	PC06
3	PC02
4	PC08
5	PC01
6	PC09
7	PC04
8	PC03
9	PD00
10	P510
11	PC07
12	PC13
13	P509
14	P508
15	P505
16	PA01
17	PA02
18	PA05
19	P611
20	P614
21	P612
22	P609
23	P610
24	P113
25	PA13
26	PA14
1	PC10
2	P607
3	PC05
4	PC11
5	PC14
6	PC12
7	PA04
8	PC03
9	P608
10	PC15
11	P506
12	P507
13	P504
14	P503
15	PA03
16	PA00
17	P301
18	P615
19	PA12
20	P613
21	P114
22	P115
23	P112
24	PA11
25	P302
26	P300

Audio Devices

PDM MEMS Microphones



SSIE/I2S Audio Codec

